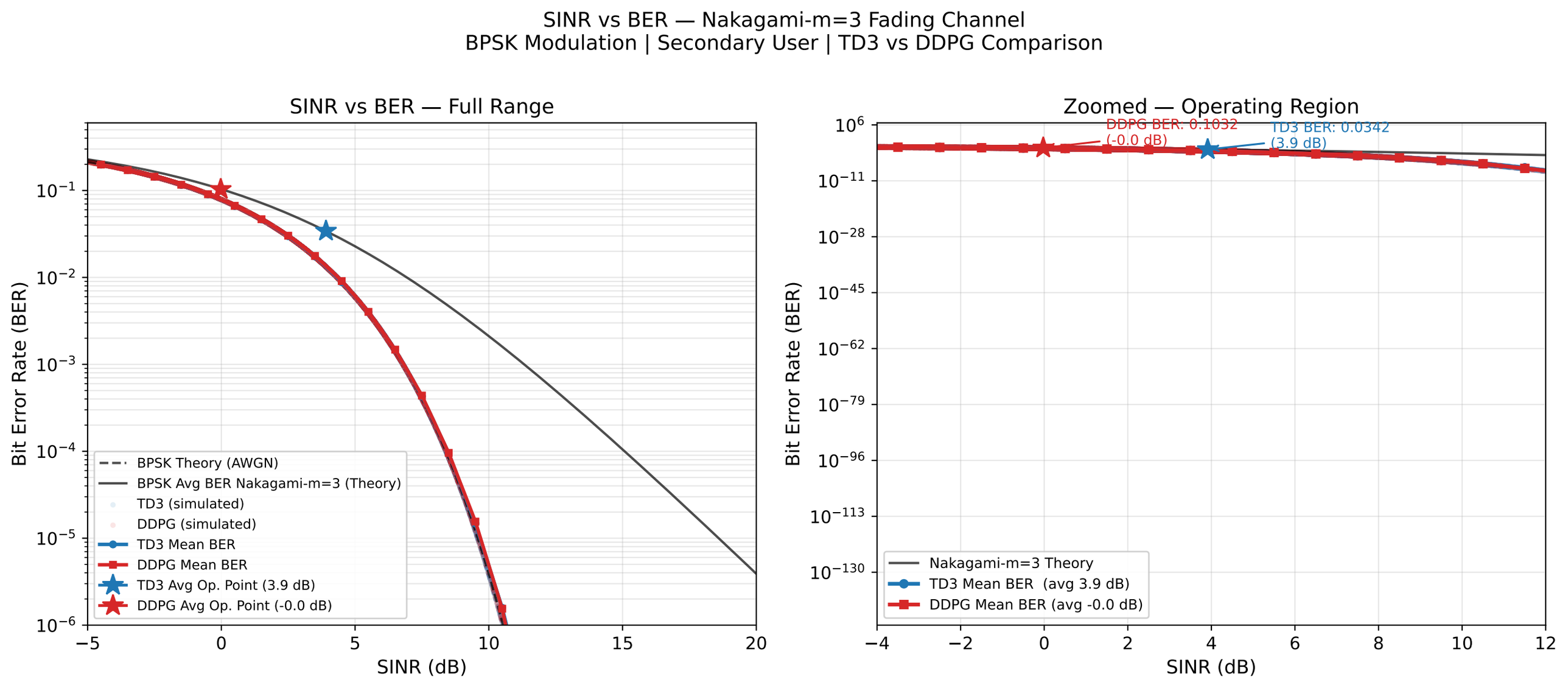
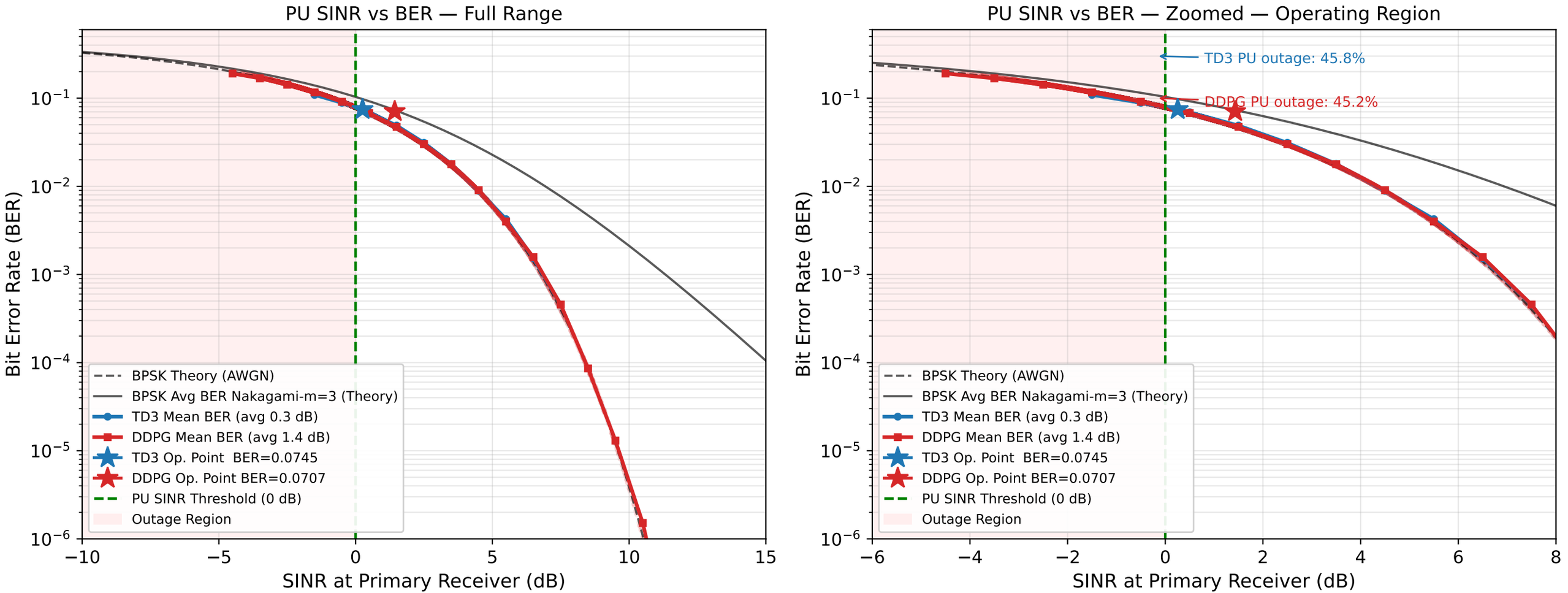
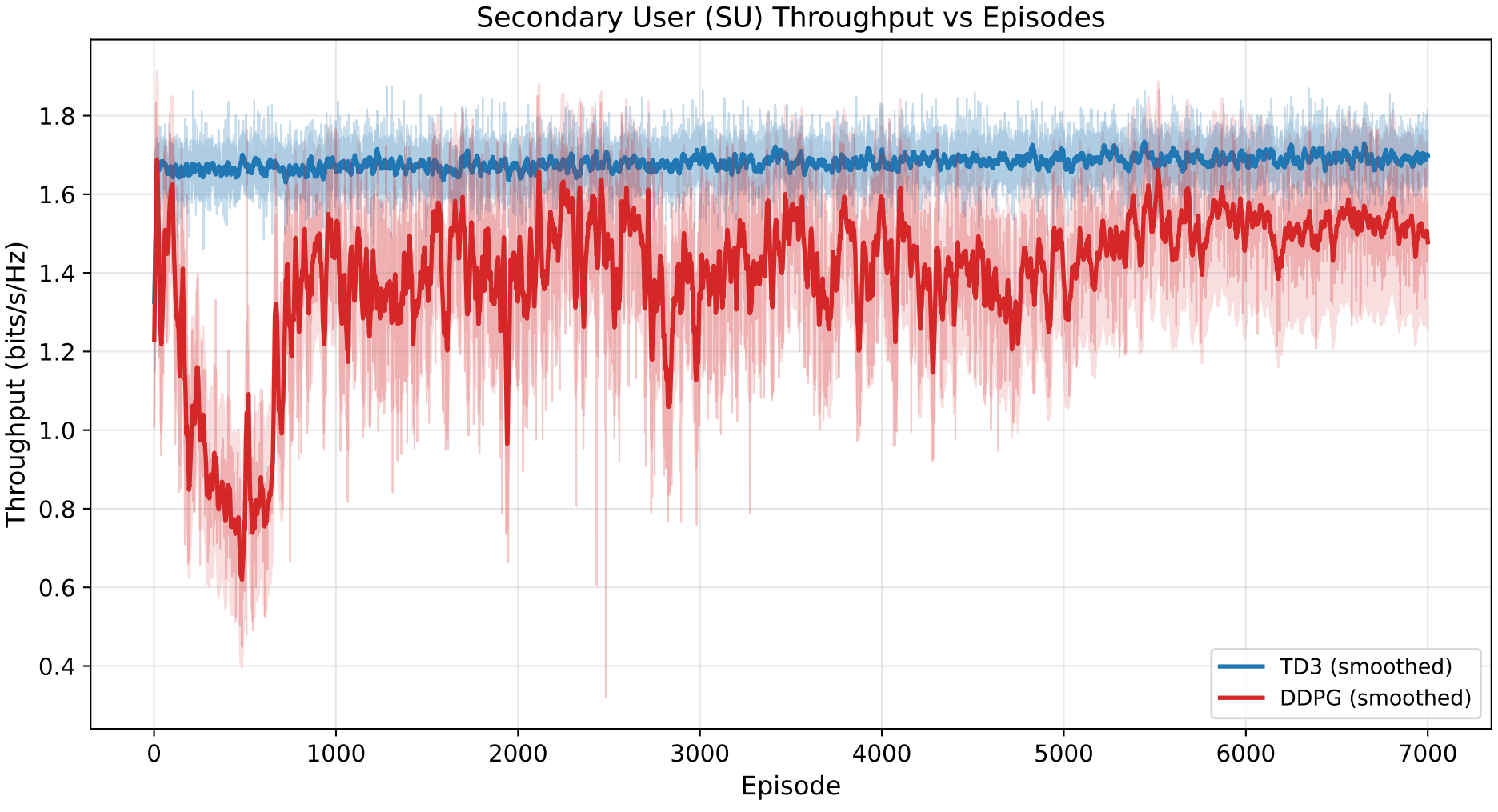
CRN Power Control: TD3 vs DDPG
Nakagami-m Fading Channel Performance Report

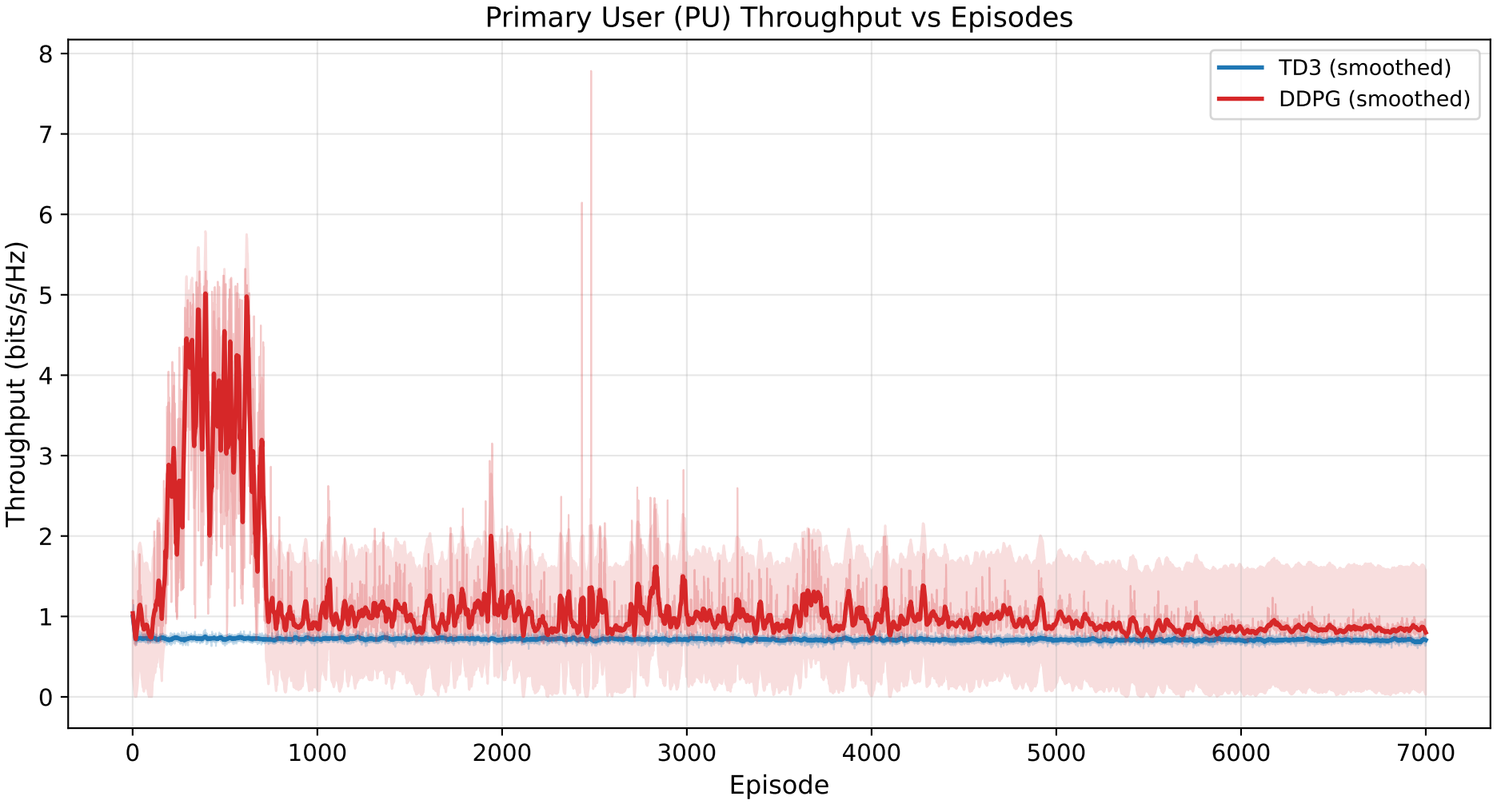
Nakagami-m = 3.0 | Episodes = 7000 | Steps/Episode = 200 | SINR Threshold = 1.0 dB

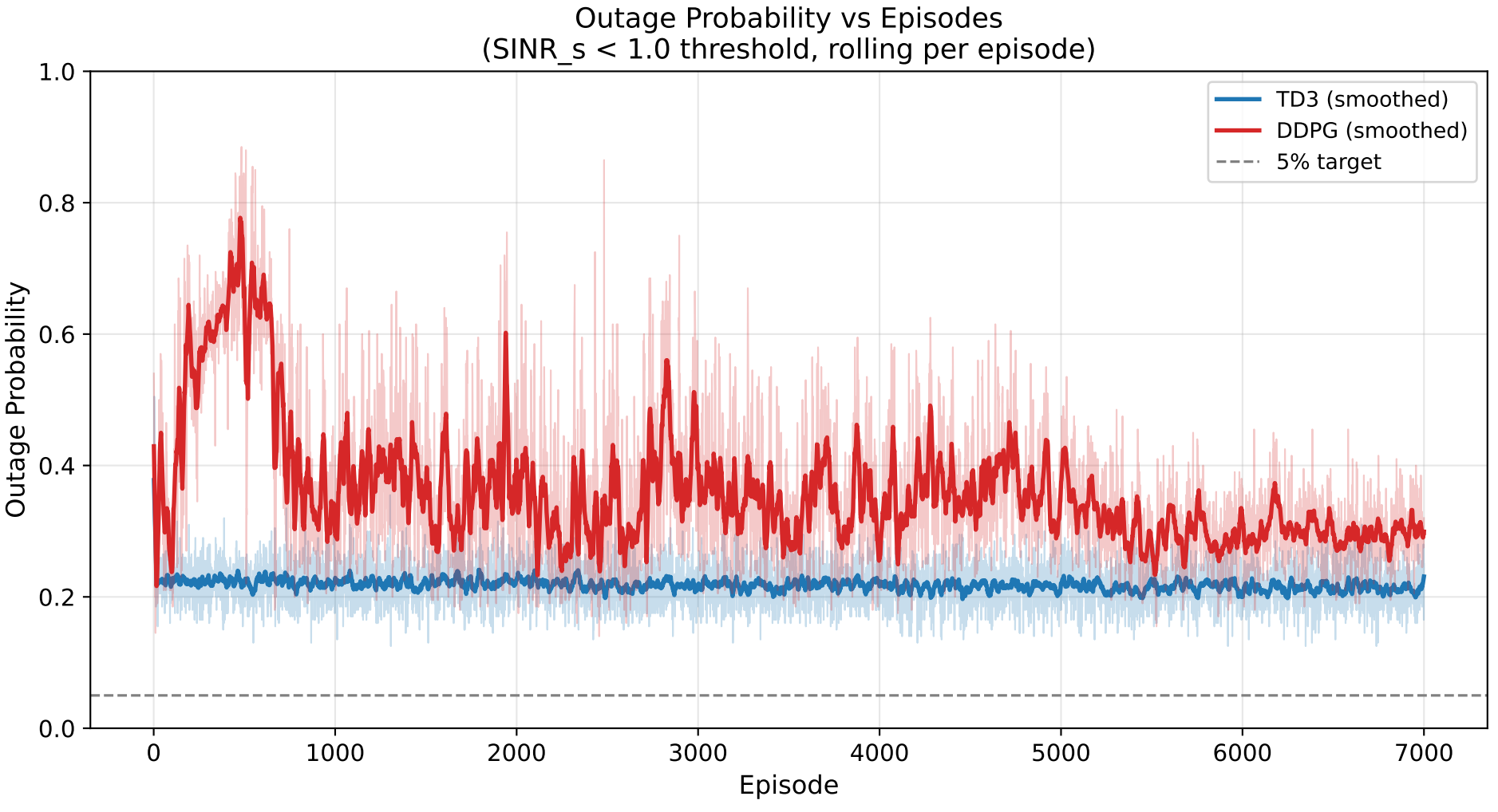
Metric	TD3	DDPG	Winner
Avg Reward (last 100 ep)	158.9453	153.8011	**TD3**
SU Throughput (bits/s/Hz)	1.6919	1.5026	**TD3**
PU Throughput (bits/s/Hz)	0.6997	0.8500	**DDPG**
Outage Probability	0.2113	0.3009	**TD3**
Average BER	0.046006	0.061741	**TD3**
Training Time (s)	10761.3	25202.7	—

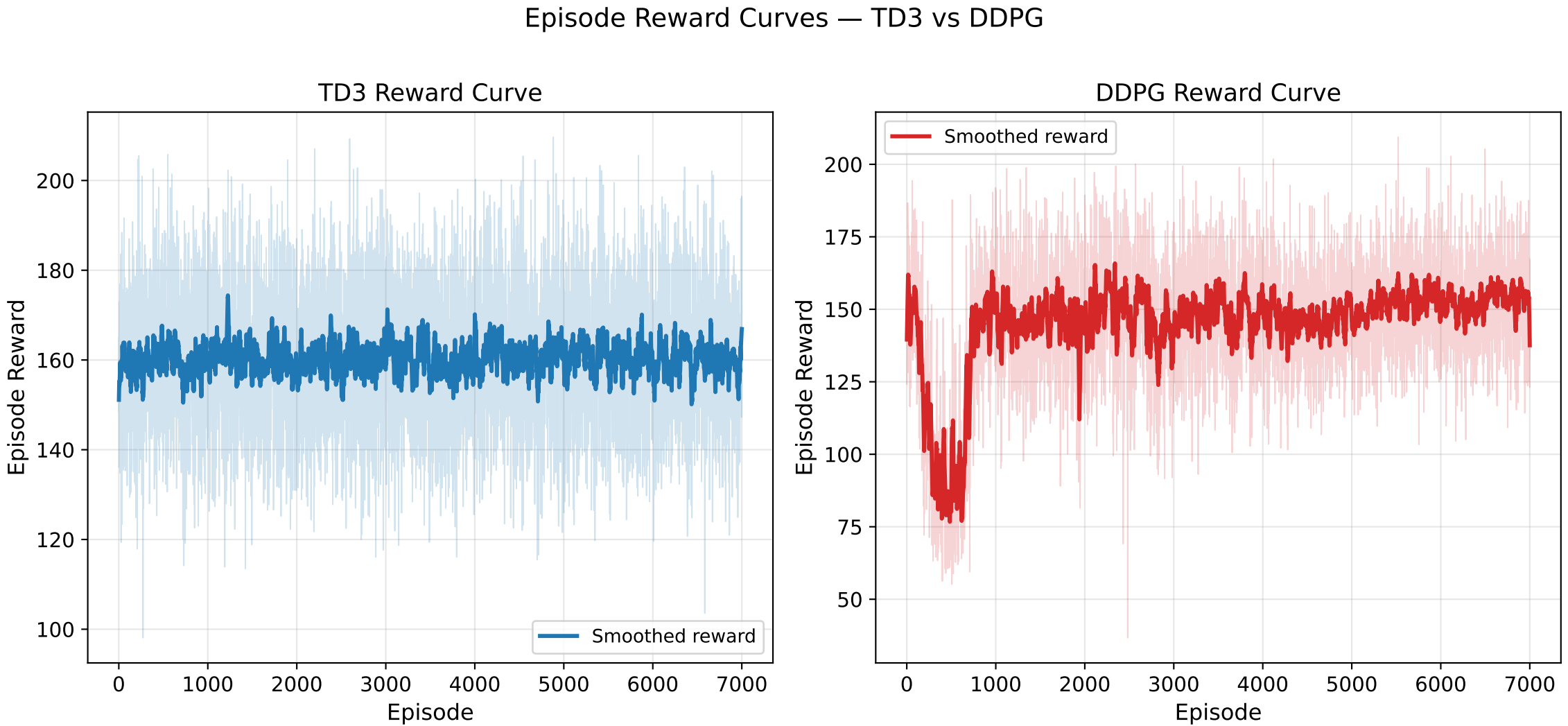

SINR vs BER — Nakagami-m=3 Fading Channel
BPSK Modulation | Secondary User | TD3 vs DDPG Comparison
SINR vs BER — Full Range
Bit Error Rate (BER)
SINR (dB)
BPSK Theory (AWGN)
BPSK Avg BER Nakagami-m=3 (Theory)
TD3 (simulated)
DDPG (simulated)
TD3 Mean BER
DDPG Mean BER
TD3 Avg Op. Point (3.9 dB)
DDPG Avg Op. Point (-0.0 dB)
Zoomed — Operating Region
DDPG BER: 0.1032
(-0.0 dB)
TD3 BER: 0.0342
(3.9 dB)
Nakagami-m=3 Theory
TD3 Mean BER (avg 3.9 dB)
DDPG Mean BER (avg -0.0 dB)

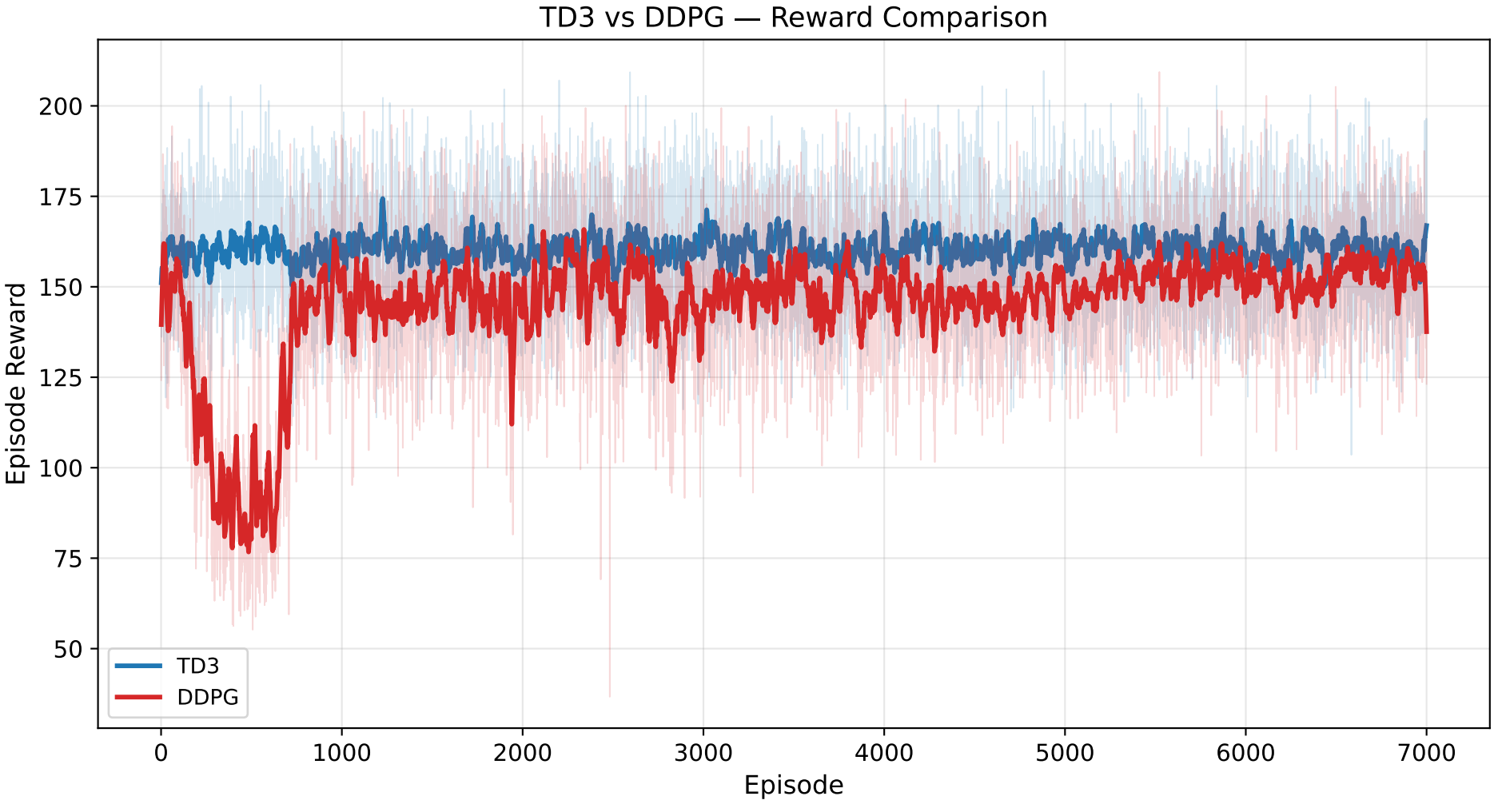

Primary User (PU) SINR vs BER — Nakagami-m=3 Fading Channel
BPSK Modulation | Effect of SU Interference under TD3 vs DDPG Power Control
PU SINR vs BER — Full Range
PU SINR vs BER — Zoomed — Operating Region
Bit Error Rate (BER)
SINR at Primary Receiver (dB)
BPSK Theory (AWGN)
BPSK Avg BER Nakagami-m=3 (Theory)
TD3 Mean BER (avg 0.3 dB)
DDPG Mean BER (avg 1.4 dB)
TD3 Op. Point BER=0.0745
DDPG Op. Point BER=0.0707
PU SINR Threshold (0 dB)
Outage Region
TD3 PU outage: 45.8%
DDPG PU outage: 45.2%


Secondary User (SU) Throughput vs Episodes
Throughput (bits/s/Hz)
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0
1000
2000
3000
4000
5000
6000
7000
Episode
TD3 (smoothed)
DDPG (smoothed)


Primary User (PU) Throughput vs Episodes
TD3 (smoothed)
DDPG (smoothed)
Throughput (bits/s/Hz)
Episode
0
1
2
3
4
5
6
7
8
0
1000
2000
3000
4000
5000
6000
7000


Outage Probability vs Episodes
(SINR_s < 1.0 threshold, rolling per episode)
TD3 (smoothed)
DDPG (smoothed)
5% target
Outage Probability
Episode
0.0
0.2
0.4
0.6
0.8
1.0
0
1000
2000
3000
4000
5000
6000
7000


Episode Reward Curves — TD3 vs DDPG
TD3 Reward Curve
Episode Reward
Episode
Smoothed reward
200
180
160
140
120
100
0
1000
2000
3000
4000
5000
6000
7000
DDPG Reward Curve
Episode Reward
Episode
Smoothed reward
200
175
150
125
100
75
50
0
1000
2000
3000
4000
5000
6000
7000


TD3 vs DDPG — Reward Comparison
Episode Reward
Episode
200
175
150
125
100
75
50
0
1000
2000
3000
4000
5000
6000
7000
TD3
DDPG